2026 MCM Senior Group Problem A
SafeInvest: Building a Trustworthy Robo-Advisor

Background and Challenge

SafeInvest is a fictional fintech company that offers investment advice through a mobile app. New customers complete a short digital questionnaire, and an automated system recommends one of three portfolios:

- Conservative,
- Balanced, or
- Growth.

Each portfolio uses equities, government bonds, and a cash-like short-term investment. After a customer invests, the app reviews the portfolio once per period and may retain the current holdings, rebalance, change the target allocation, or request review by a human adviser.

Automation can make advice inexpensive and widely available, but it also creates new risks. A portfolio may be unsuitable for a customer, an algorithm may trade too often, and a recommendation may be difficult to explain. The system must also recognize conditions in which its model should not be trusted.

In this problem, we want to develop and test an automated, explainable, and risk-aware robo-advisory process for SafeInvest. The process must recommend suitable portfolios, adapt them only when the expected benefit justifies the cost, and identify decisions that require human review.

Customer Information

SafeInvest's questionnaire records investment horizon, the largest temporary loss in purchasing power that the customer believes they could tolerate, and liquidity need. Five representative responses are shown below. Loss tolerance is a customer response, not a guaranteed portfolio limit.

Profile	Horizon	Loss tolerance	Liquidity need
A	5 years	10%	High
B	15 years	25%	Medium
C	30 years	40%	Low
D	25 years	15%	Medium
E	6 years	35%	High

Provided Market Data

The file `SafeInvest_Synthetic_Data.csv` contains 180 independent synthetic market scenarios, each lasting 30 annual periods.

- *scenario_id*: Scenario number, 1–180;
- *period* : Annual period within the scenario, 1–30;
- *equity_return* : Nominal total equity return;
- *bond_return* : Nominal total government-bond return;
- *bill_proxy_return* : Nominal short-term bill proxy return;
- $inflation$: Inflation during the period.

Returns and inflation are decimals. For example, 0.05 means 5%. For nominal return r_t and inflation π_t, real return is

$$r_t^{\text{real}} = \frac{1 + r_t}{1 + \pi_t} - 1.$$

The data are synthetic and are based on historical U.S. market data. They are not historical events or forecasts.

Common Conditions

Every market scenario begins with 100 units in bills and has no later deposits or withdrawals. Portfolios must be long-only and fully invested. SafeInvest may review a portfolio once per period.

Charge 0.25% of the money moved between assets. Count each amount transferred between assets only once when calculating transaction costs. Moving 20 units from bills to equities therefore costs 0.05 units. Deduct the transaction cost from portfolio wealth at the time of trading, before applying that period's returns. Charge the initial allocation from bills in the same way. Ignore taxes, other fees, and terminal liquidation costs.

A decision for period t may use only observations through period $t-1$ of that scenario. It may not use period t or any later observation from that scenario.

Questions

Address the following tasks as one coherent investigation.

1. Design the digital suitability process. Choose measures of investment growth and risk. Express the large-loss measure so that larger values mean greater loss. Your risk assessment must separate ordinary fluctuations from unusually large losses and must account for purchasing power.

Recommend allocations or allocation rules for the Conservative, Balanced, and Growth portfolios. Then design a transparent onboarding rule that uses the questionnaire information to recommend one of the three portfolios or request human review. Apply the rule to Profiles A–E with a brief explanation.

2. Design and evaluate the automated portfolio policy. For a customer assigned to the Balanced portfolio, develop a rule that decides whether to keep the current holdings, rebalance, change the target allocation, or request human review. Trade or change the target only when the estimated benefit justifies the transaction cost.

Use an equal-weight baseline whose target remains one-third in each asset. Restore these weights at every review and charge the stated cost. Compare your policy with this baseline across the supplied scenarios. For each strategy, report the median growth and large-loss results, the 10th percentile of growth, and the 90th percentile of the large-loss measure. Briefly examine one severe-loss scenario and report the percentage of review periods in which your policy trades. Decide whether automated weight updates add value.

Include one customer-facing explanation of no more than 50 words for a trade, target change, or request for human review.

3. Test reliability and define safeguards. Test one important modeling assumption or the transaction-cost rate and state whether your main recommendation changes. Identify one important limitation and one condition that should trigger human review or suspension of the automated policy. State what the app should do while review is pending.

Conclusion. Give a concise recommendation that defines the three portfolios, assigns Profiles A–E, states whether the Balanced portfolio should use automated weight updates, and identifies the main safeguard. When evaluating recommendations for Profiles A–E, take their stated investment horizons into account. You may use Scenarios 1–120 to develop and adjust your model. Evaluate your final policy using Scenarios 121–180 without further adjustment.

Instructions

- **Use of AI Tools**: Teams are allowed to use AI tools, including ChatGPT, DeepSeek, and other AI-assisted software. However, all AI use must be clearly disclosed. In the report, please include a brief paragraph on *Use of AI Tools* at the end of your submission, including the tools used and their purposes. All AI-generated content, calculations, code, and references must be carefully checked for correctness and plagiarism. Teams remain fully responsible for their submitted work.
- **Submission**: Each team should submit a short report (**no more than 10 A4 pages, font size no smaller than 11pt**) in English explaining your answers and ideas. Please include the most important figures, tables, mathematical models, and numerical results in the report.

  Please list the names of all members in your report. Each team only has to ask the Team Leader to submit the files through the CMT system before October 20, 2026, 23:59 Beijing Time:

  `https://cmt3.research.microsoft.com/HKUSTGZMCM2026`

  – Main submission file: The report, in doc, docx, or pdf format.
  – If there are any source codes used, please submit them as **Supplementary Materials** in a **ZIP** file no larger than 50 MB.
- **Copyright:** Each team retains the copyright of their work. We will only use your submissions to decide the winner.
- **Data Source and Acknowledgment**: The SafeInvest Synthetic Dataset is based on selected United States series from the Jordà–Schularick–Taylor Macrohistory Database, Release 6. It does not reproduce the historical observations. The SafeInvest Synthetic Dataset and accompanying problem statement are distributed for noncommercial use under the license CC BY-NC-SA 4.0. Redistribution or adaptation must retain the attribution, identify changes, and use the same or a compatible license.
- **Academic Integrity:** This competition, like all other competitions, functions best when the participants treat one another with honesty, fairness, respect, and trust. The organizers hold all participants of this competition to high standards of scholarship and integrity.

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